Quadratic-phase leakage of the small-x factor base (lead N2)

Lead: LEAD-EXO-quadratic-phase-small-x-leakage. Protocol protocol.json, QUADPHASE-20261104, was frozen before data (sha256 6a44309417a9..., commit 2d6c661a), with the run and analysis scripts pinned by hash.

Verdict (frozen rule): NULL, with valid controls. At $N \approx 2^{12-2}14$, the small-x factor base's discrete logs correlate with quadratic phases $e((\alpha l^2 + \beta l)/N)$ no more than negation-closed random sets of the same size do. This closes degree-2 phase leakage of the small-x predicate at toy scale, beside H-PSEUDO's linear phases, and it carries the same scale caveat: a toy size cannot see a correlation of order 1/polylog.

Design

- **Statistic.** $Q(S) = \max$ over $\alpha \in [1, N-1]$ and $\beta \in [0, N-1]$ of $|\Sigma_{l \in S} e((\alpha l^2 + \beta l)/N)| / \sqrt{|S|}$, computed with one length-N FFT per α.
- **Curves.** Each has prime order $N \neq p$, with p the largest prime below 2^L. There are 3 curves at each decisive size L = 12, 13, 14, and 1 descriptive curve at L = 16.
- **Arms.**
  - EC: the discrete logs of the first 512 points sorted by (x, y), the H-PSEUDO small-x rule.
  - Null: 40 negation-closed random 512-sets per curve (5 at L = 16), applying the D1 lesson.
  - Planted: the negation-closed set $\{l : \alpha_0 l^2 \bmod N < 512\}$, which must spike at α_0.
- **Rule.**
  - SIGNAL iff Q_EC exceeds every null draw on 2 of 3 curves at 2 of the 3 decisive sizes.
  - NULL iff Q_EC's percentile lies in [0.05, 0.95] on ≥ 8 of 9 decisive curves.
  - Validity requires planted Q ≥ 3 × null max with the argmax at α_0, and A/A within 3%.
- **Sizes.** These were reduced from the memo's 2^14 and 2^16, because one Q evaluation costs about 400 s at 2^16 here. That reduction is recorded in the protocol.

Results

size	N	Q_EC	null mean / max	u	planted / null max
2^12	4133	4.102	4.028 / 4.512	0.650	4.9×
2^12	3989	4.144	3.990 / 4.642	0.700	4.7×
2^12	4007	4.268	4.036 / 4.787	0.850	4.8×
2^13	8231	4.152	4.224 / 4.658	0.375	5.2×
2^13	8081	4.151	4.257 / 4.995	0.375	4.4×
2^13	8167	3.997	4.187 / 4.928	0.175	4.6×
2^14	16417	4.303	4.415 / 5.400	0.350	4.0×
2^14	16273	4.579	4.524 / 5.678	0.625	3.8×
2^14	16427	4.201	4.405 / 4.894	0.125	4.5×
2^16	65167	4.725	4.609 / 4.708	1.000	4.8×

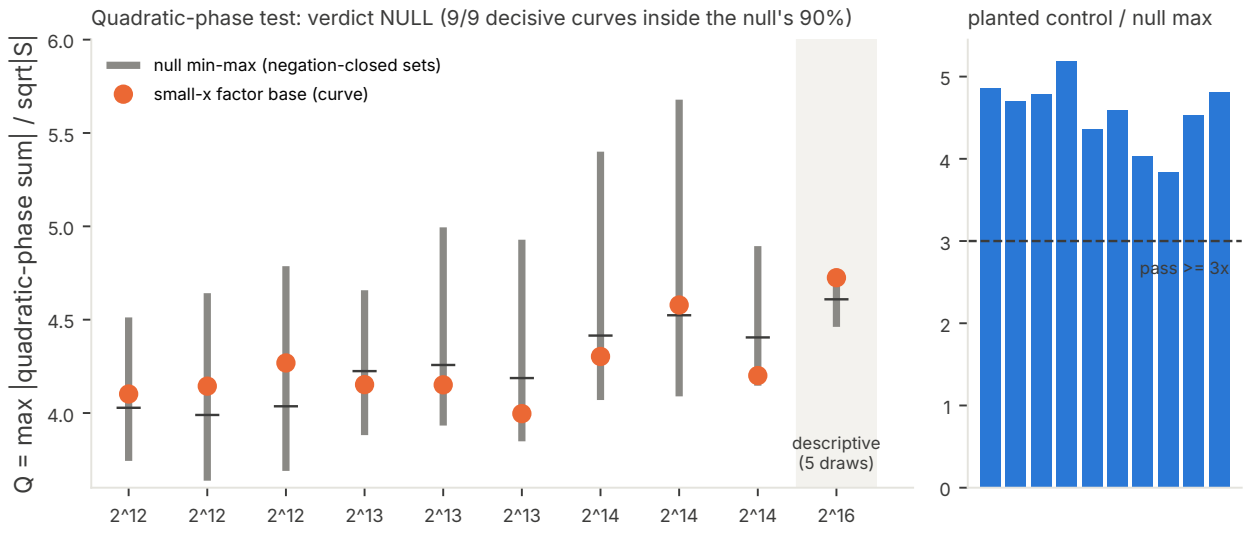

Figure 1: Q per curve

- **All decisive curves are inside the null's band.** 9 of 9 decisive curves fall inside it (u from 0.125 to 0.85), and no size has a curve above the null maximum.
- **The controls are valid.** The planted control is 3.8–5.2× the null maximum on every curve, always at the planted α_0, and the A/A even/odd difference is at most 2.3%.
- **2^16 is descriptive only.** That one curve sits above all 5 of its null draws (Q 4.725 against max 4.708). With 5 draws this happens with probability 1/6 under the null. It is recorded, not interpreted. A 2^16 rerun with 40 draws would settle it, at about 4.5 core-hours here.

Consequence

The lead's discriminator is met on the NULL side at toy scale. Linear (H-PSEUDO) and quadratic phases of the small-x predicate in log order both look like those of a random negation-closed set at these sizes. The exploitation path in the lead (sparse Fourier recovery of k^2) has nothing to act on there.

Deviations

- No S3 checkpoint (no upload token). With the owner's approval, the pushed branch is the resume record.
- The run shared the cores with three other runs. Values are exact, so only wall time was affected.

Reproduce

```
xz -dc rows.jsonl.xz > rows.jsonl     # or rerun:
python3 quadphase.py --protocol-sha 6a44309417a905167606a10754a6fad66de617a16c81feed73cd9cace0a02efb --out rows.jsonl --workers 3
python3 analyze_quadphase.py rows.jsonl verdict.json
python3 figures.py; ./build_pdf.sh
```